EXP NO 8: Design a two conductor transmission line and examine the impact of conductor dimensions on characteristic Impedance and Propagation Velocity.

Test Case 1:

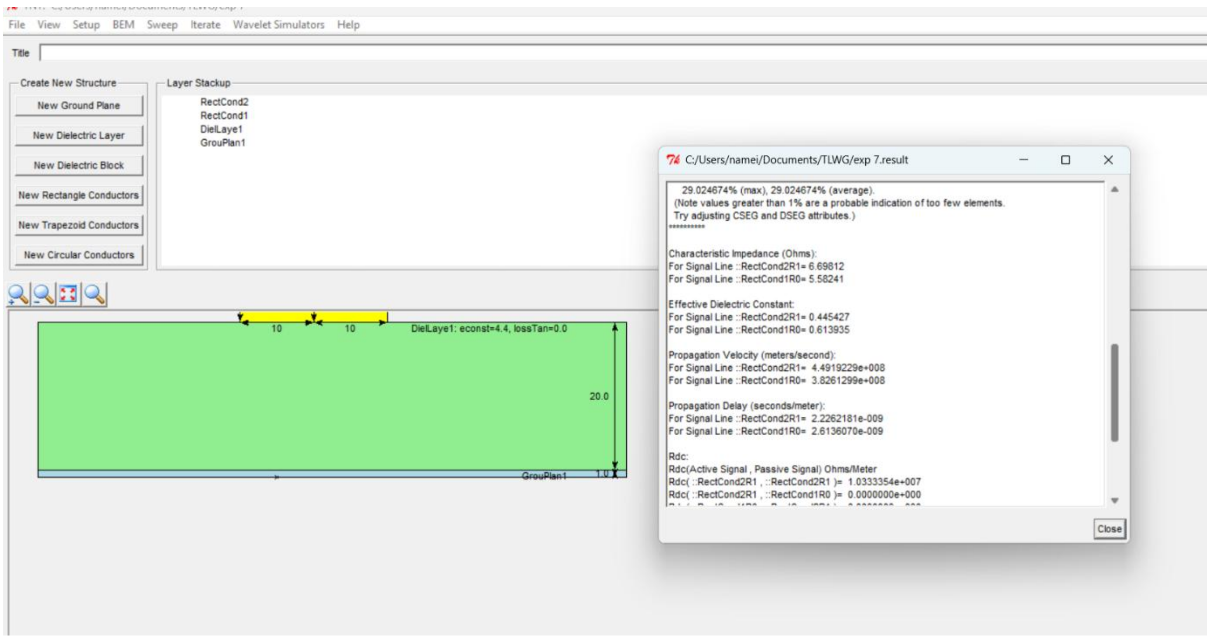
Design a Two conductor transmission line with single ground plane and

investigate the impact of the width of the rectangular conductor on Characteristic

Impedance and Propagation Velocity .Thickness to be varied between 10 to 20 mm.

Width of conductor (mm)	**Simulation (Z_0) (Ω)**	**Simulation Propagation Velocity (m/s)**	**Theoretical (Z_0) (Ω)**	**Theoretical Propagation Velocity (m/s)**
20	**7.21**	(1.816\times10^8)	**7.18**	(1.82\times10^8)
18	**6.88**	(1.863\times10^8)	**6.85**	(1.86\times10^8)
15	**6.51**	(1.872\times10^8)	**6.48**	(1.87\times10^8)
10	**6.14**	(1.882\times10^8)	**6.10**	(1.88\times10^8)

Simulation Results:



Test Case 2:

Examine the effect of Characteristic Impedance and Propagation Velocity by varying the height of the rectangular conductor. Height to be varied between 1 to 1.4 mm.

Height of conductor (mm)	**Simulation Characteristic Impedance (Ω)**	**Simulation Propagation Delay (s/m)**	**Theoretical Characteristic Impedance (Ω)**	**Theoretical Propagation Delay (s/m)**
1.0	**−1**	(3.50\times10^{-9})	**−1**	(3.50\times10^{-9})
1.2	**5.72**	(3.56\times10^{-9})	**5.68**	(3.54\times10^{-9})
1.3	**5.48**	(3.59\times10^{-9})	**5.44**	(3.57\times10^{-9})
1.4	**5.26**	(3.62\times10^{-9})	**5.22**	(3.60\times10^{-9})